

CustoMate — Complete Change Log

A Customer-Guided 3D Customization Platform with Order Management for Bryle Closet Printing Services. Tracks every system-side fix against the May 20, 2026 reoral defense compliance form, plus the design-preview lifecycle, role-system redesign, and customizer polish shipped in this iteration.

1. Compliance Form Recommendations

Prof. Emil Karlo E. Flores (Panel Chair)

#	Recommendation	Status	Implementation
1	Color theme should be more lively	DONE	Cinematic navy/aurora studio backdrop, gradient sidebar tabs, brand-tinted icons.
2	3D render shows ideal model per product	DONE	AI Lifestyle Preview with body-size presets; 6 product types supported.
3	Clarify order processing	DONE	Status enum + audit log; inventory consumption on approval; OrderTracking timeline.
4	Calendar efficiency	DONE	Production calendar with span highlighting, drop-to-schedule, pending strip.
5	Demonstrate AI Capability	DONE	Gemini text (decals) + image (designs + lifestyle) + Admin AI Assistant.
6	Finalize complete flow	DONE	End-to-end: design → cart → checkout → payment → fulfillment.
7	Custom presets	DONE	Templates Panel with 10 designs across 5 categories.

Prof. Patrick Nicole C. Ramos

#	Recommendation	Status	Implementation
1	Register new user fix	DONE	Brevo HTTPS OTP on port 443 (Render blocks SMTP).
2	Design on different products	DONE	3D customizer works on 6 product types.
3	Calendar shows progress	DONE	Start / Due / Day N/M badges + mini progress bars per span.
4	Account security (users ≠ admin)	DONE	adminMiddleware guards every admin route + role-check on JWT.
5	Inventory validations + audit trail	DONE	Append-only StockMovement schema + InventoryAuditLogModal + manual SKU.
6	Reports fix	DONE	analytics.js totalPrice/status enum fix + PDF export.
7	AI integration on Design	DONE	AIDesignAssistant in customizer (Gemini decal gen with auto-sticker).
8	Appealing customer UI	DONE	Checkout redesign, sidebar tabs, smart-sticker banner.

Prof. Cristine Erica E. Untalan

#	Recommendation	Status	Implementation
1	Email OTP	DONE	Brevo HTTPS API; SMS OTP removed.

#	Recommendation	Status	Implementation
2	Calendar functions	DONE	Drop-to-schedule, span highlighting, bulk scheduling.
3	Update UI color	DONE	Consistent palette across customer + admin.
4	Rush orders	DONE	Urgency tier badges + rush fee in checkout.
5	Multi-day schedule (5-6 days)	DONE	productionDate → productionDueDate span rendering.
6	Start & end display	DONE	Start badge (blue), Due badge (emerald), Day N/M chip on intermediates.
7	Export PDF on every table	DONE	Orders, Inventory, Users, Coupons, Reports + Audit Log all export PDF.
8	3D works on all products	DONE	All 6 product types render in 3D.
9	Inventory log + manual SKU	DONE	Manual SKU input + uniqueness validation; audit log tags actor + role.

2. Customization Studio (Core Feature) Polish

Feature	Status	Notes
Cinematic studio backdrop	DONE	Navy + aurora blobs, 9-second pulse, dot-grid texture, spotlight halo.
Templates Panel	DONE	10 presets across 5 categories (Sports / Minimal / Bold / Fun / Vintage).
Quick Shapes panel	DONE	12 shapes at 3000px print resolution.
ImageRefineModal	DONE	Background removal, crop, edge feather, contrast, saturation, Auto-Sticker preset.
Auto-Stickerize utility	DONE	Runs on every upload + AI generation. Safe fallback on photos / transparent PNGs.
Smart Sticker Mode banner	DONE	Violet/fuchsia/pink hero in Image tab.
Brand-tinted sidebar tabs	DONE	Text (blue/indigo), Image (violet/fuchsia), AI (fuchsia/pink), Options (slate).
Mirror / Flip image	DONE	H + V flip buttons.
Text effects	DONE	Stroke, drop shadow, letter spacing, preset chips (Stadium / Soft / Clean).
Opacity control	DONE	10-100% slider on image decal.
Undo / Redo + keyboard shortcuts	DONE	Ctrl+Z/Y, arrow nudge (Shift = 5%), +/- scale, R rotate, Delete; help via ? button.
Snapshot + Share buttons	DONE	High-res PNG download + clipboard / native-share.
Recently-used colors palette	DONE	localStorage-persistent; swatch row under every color picker.
AI prompt improved	DONE	Gemini asked for die-cut stickers on PURE white background.
Image quality: alphaTest 0.05	DONE	Soft feathered edges survive without square halo.
MeshBasicMaterial (unlit)	DONE	Pixel-perfect color fidelity (no ACES tone-mapping wash).
Front/back duplication fix	DONE	filterDecalByFacing strips back-facing triangles from DecalGeometry.
Multi-layer support	DONE	LayersPanel + Text / + Image buttons; non-default layers preserved.
Page locked to viewport	DONE	h-screen + overflow-hidden so the page itself never scrolls.
Tighter sidebar density	DONE	space-y-6 → space-y-4, padding p-6 → p-5.
Print-quality analyzer	DONE	DPI check against physical print size for the chosen product.

3. Design Preview → Order Lifecycle

Step	Status	Notes
Studio captures 3D canvas snapshot on Add to Cart	DONE	~113KB PNG dataURL stored on the cart item.
Cart shows design thumbnail + Custom badge	DONE	Falls back to product photo when no preview.

Step	Status	Notes
Checkout forwards previewImage + designConfig	DONE	Was being dropped before; now wired through.
Order schema persists previewImage + designConfig	DONE	Optional Cloudinary upload swaps dataURL for hosted URL.
OrderTracking shows thumbnail + Open + Download	DONE	One-click download with order-id-derived filename.
Admin OrderDetailDrawer preview + dual download	DONE	"Preview PNG" + "Artwork PNG" buttons for production team.

4. Backend Infrastructure

Item	Status	Notes
Brevo HTTPS API for OTP send	DONE	Replaces Gmail SMTP — Render blocks ports 465 + 587.
Unified mailer service	DONE	Brevo → Resend → SMTP fallback chain.
MAIL_FROM env var support	DONE	Sender match against verified Brevo sender.
MongoDB Atlas IP whitelist	DONE	0.0.0.0/0 + specific IPs.
PayMongo webhook HMAC verification	DONE	Already in place.
Order inventoryConsumed flag	DONE	Prevents double-deduct on status transitions.
Stock consumption on order approval	DONE	First transition into approved fires consumeReservedForOrder.
StockMovement audit logging	DONE	Every restock / sale / adjustment / damage / release tagged with actor name + role.
ProductionLog enum includes “approved”	DONE	Fixes "Server error" on Schedule modal.
Production schedule span query	DONE	Overlap-based (was start-day only).
Analytics totalPrice fix	DONE	Reports now show real revenue.
Status enum normalisation	DONE	pending / approved / in_production / ready / completed / shipped / delivered / cancelled / rejected / refunded.
Backend image gen prompt	DONE	Gemini asked for sticker on white background, hard edges, single subject.

5. Admin Panel Additions

Item	Status	Notes
Admin Reports PDF + CSV export	DONE	@media print stylesheet, A4 layout.
AdminOrders PDF export	DONE	Via shared PrintablePage component.
AdminInventory PDF export	DONE	Plus Audit Log + Suppliers buttons in hero.
AdminUsers PDF export	DONE	Beside CSV export.
AdminCoupons PDF export	DONE	Beside CSV export.
InventoryAuditLogModal	DONE	Global cross-SKU ledger with type/date filters + CSV + PDF.
Manual SKU on inventory create	DONE	Frontend input + backend uniqueness check.
Admin AI Assistant model fix	DONE	gemini-2.0-flash → gemini-2.5-flash (9 occurrences).
Admin Calendar production spans	DONE	Multi-day blue blocks with Start / Due / Day N/M badges.
Admin Calendar pending strip	DONE	"Orders waiting to be scheduled" chips + drop-to-day.
Admin Calendar Schedule button per queue item	DONE	Opens ScheduleOrderModal.

6. Role System (Two Personas)

Item	Status	Notes
User.role enum	DONE	customer / admin (= Production Manager) / guest / production_staff.
requireRoles(...) factory	DONE	Generic allow-list with 403 + accepted-roles in body.
requireManager alias	DONE	Aliased to adminMiddleware (Production Manager only).
requireProductionStaff alias	DONE	admin + production_staff.
sanitizeOrderForRole	DONE	Strips PII + pricing + payment for staff; admin sees full record.
jsonForRole wrapper	DONE	Applied to every production endpoint that returns orders.
Production routes role-gated	DONE	queue / schedule / active / history → staff+; stats / team / scheduling / capacity → admin.
Inventory routes role-gated	DONE	GET → staff+; POST / PUT / DELETE → admin only.
Stock movements role-gated	DONE	reads → staff+; restock / adjust / damage → admin.
Payment / coupons / users / reports	DONE	adminMiddleware (Production Manager only).
ProtectedRoute accepts role arrays	DONE	string OR string[]; admin implicitly allowed.
Per-page route gates	DONE	Orders / Reviews / Coupons / Users / Reports admin-only; Production / Calendar / Inventory admin+staff.
AdminLayout sidebar by role	DONE	Staff sees only Production + Calendar + Inventory; brand label switches.

Item	Status	Notes
AdminInventory action gating	DONE	New Product / Edit / Delete / Restock / Adjust / Suppliers / Export hidden for staff.
OrderDetailDrawer Refund gating	DONE	Hidden for non-admin.
Brand label "Production Manager"	DONE	Sidebar shows "Production Manager" or "Production Floor".
Role pill on user card	DONE	Always shows current role label.
Seed accounts live in DB	DONE	manager@ / manager123 (admin) + production.staff@ / staff123 (staff).
Migration script	DONE	Auto-promotes any leftover production_manager users to admin.

7. Deployment Artifacts

File	Size	Includes
customate-frontend.zip	199 MB	Full app: index.html, .htaccess, assets, models, oversized-t-shirt, products.
customate-product-assets.zip	46 MB	Just products + 3D models (no JS bundles).
Backend on Render	Auto-deploys	Every push to main triggers redeploy.
Frontend on Hostinger	Manual	Re-upload zip after each frontend change.

8. Commit History (this iteration)

Hash	Subject
5516419e	Collapse production_manager into admin
9b01bc34	Production Staff + Production Manager roles (initial 3-tier)
a58a61f3	Wire design preview through cart → order → tracking + admin drawer
8120675f	Compliance fixes: PDF export on every admin table + manual SKU
db2a2233	Ship .htaccess via public/ for Vite auto-copy
675651f1	Fix shape-decal print-quality block + tighten sidebar density
f57f8cc5	Studio layout: lock to viewport + bold cinematic studio backdrop
5fe1bb4d	Replace plain slate backdrop with product-photography studio
9dccb206	Auto-stickerize AI + uploads + cleaner studio sidebar UI
a42fbf85	Templates gallery + multi-layer + front/back duplication fix
e6caf0e4	Customizer polish: 12-shape library + persistent recent colors
b4bc6a3f	Add in-studio image refiner: bg removal + crop + feather
7bc14102	Switch OTP send to Brevo HTTPS API
2717a8a5	All SMTP to port 587 + STARTTLS
aadf25e9	OTP send: explicit SMTP timeouts
6568aa7d	Fix production scheduling 500 + paint full span on calendar
326c5260	Deduct stock on order approval + global inventory audit log
f6cc97cf	Fix Admin AI model name + redesign Checkout page UI
8e8eab08	Wire Production Hub to pending orders + fix Reports data accuracy

9. Test Accounts (live)

Email	Password	Role
manager@customate.com	manager123	Production Manager (admin)

Email	Password	Role
admin@customate.com	admin123	Production Manager (admin) — original
production.staff@customate.com	staff123	Production Staff

10. Outstanding Items

None on the compliance form. All system-side panelist recommendations are addressed and verified live in the running dev servers.

Generated automatically from the project change log.